



Accelerating Transparency in Home & Community Energy Management Through Benchmarking Platforms

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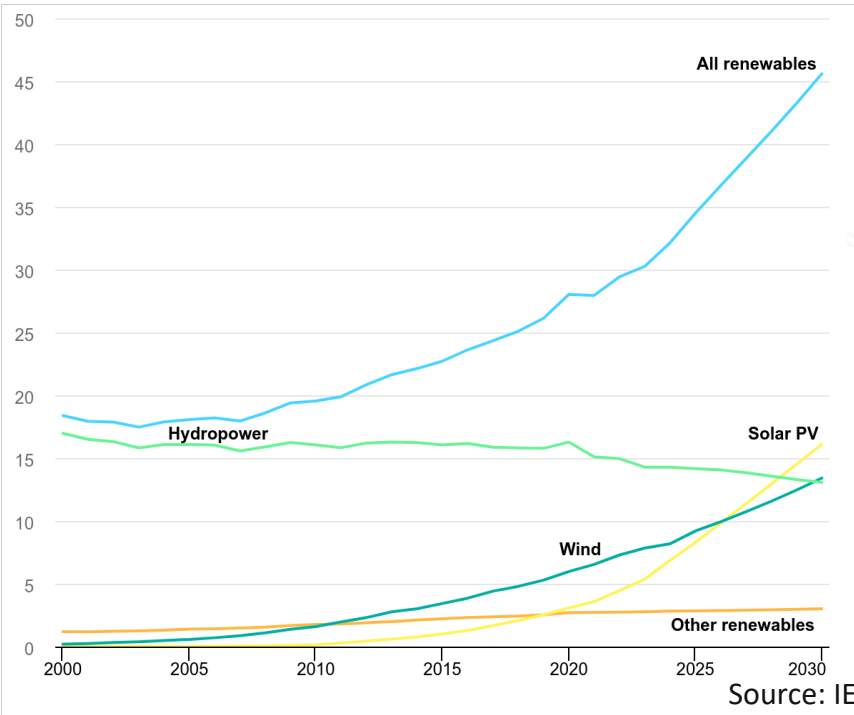
EcoFlow Technology Inc.

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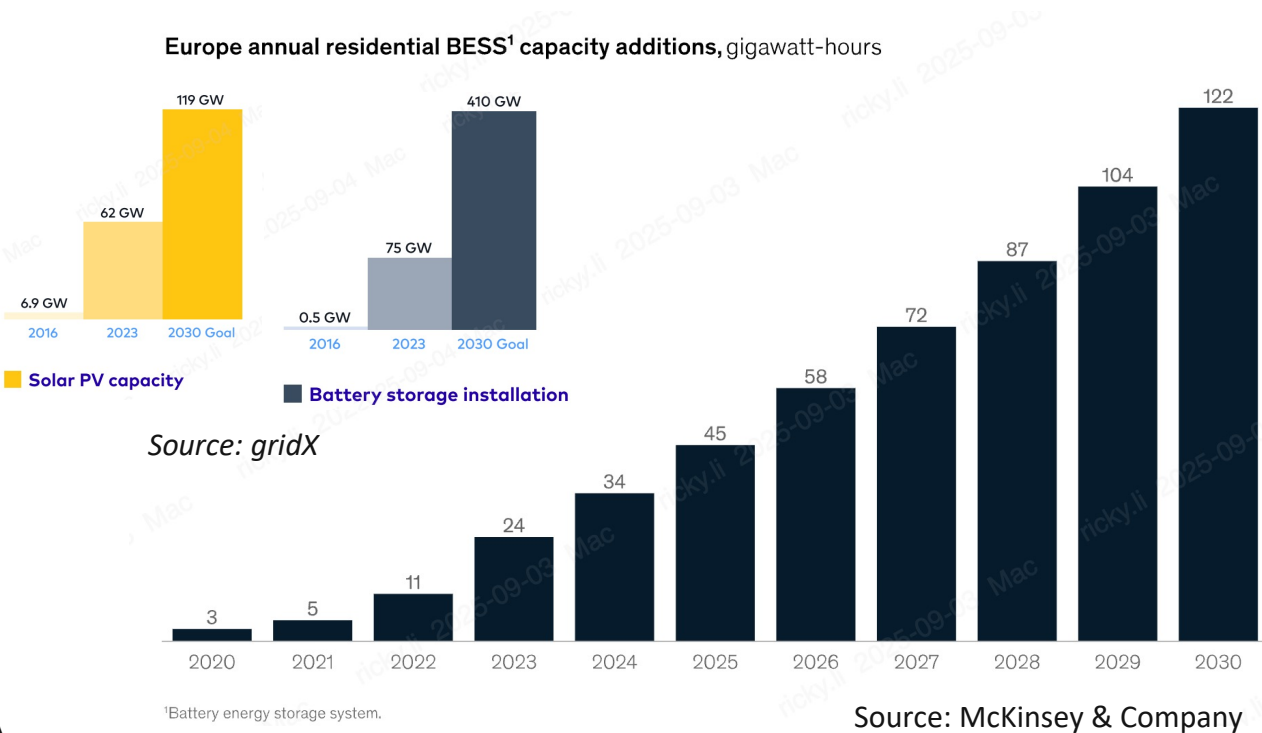


- Background
- Framework & Method
- Preliminary Result
- Demo
- User Adoption Plan
- Committer & Contributor Growth

Household installation of PV and ESS is increasingly adopted to lower electricity costs and enhance backup power availability.



By 2030, solar PV is projected to be the largest renewable energy source, contributing ~80% of global capacity growth from 2024–2030, driven by utility-scale projects and rising rooftop adoption.

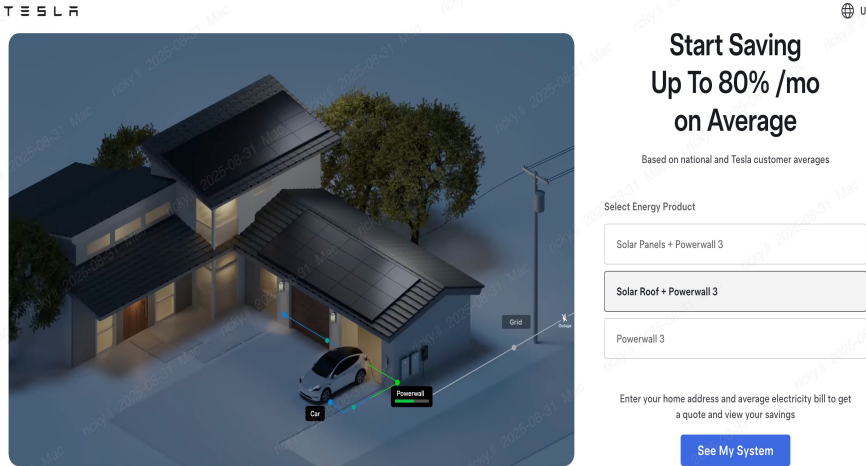


Residential BESS is projected to expand rapidly, supported by falling costs, higher electricity prices, and stronger government incentives, enabling wider integration with PV systems through 2030.

DERs (PV & ESS) location and sizing, HEMS, and grid support are key to maximizing PV/ESS revenues, but fragmented benchmarking and low transparency can yield divergent and unreliable results for end users.

1

Location & Sizing



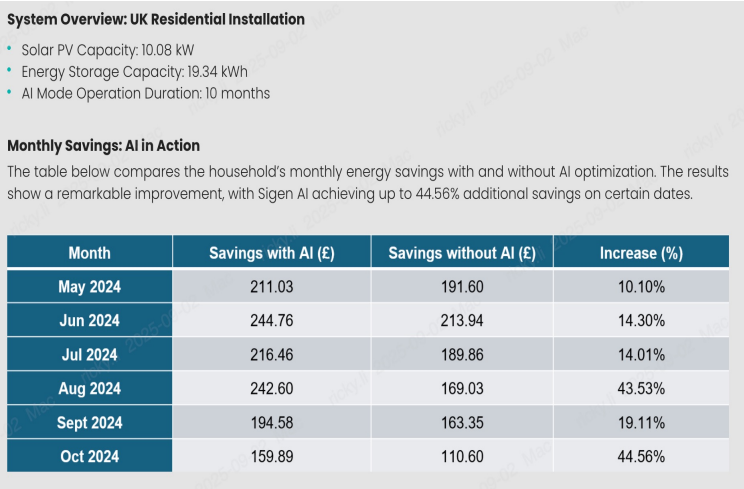
Source: Tesla

Tesla, Huawei, and other leading companies have developed advanced tools to optimize PV and ESS sizing under specified conditions.

Results vary across platforms due to differing technical methods and business considerations.

2

HEMS



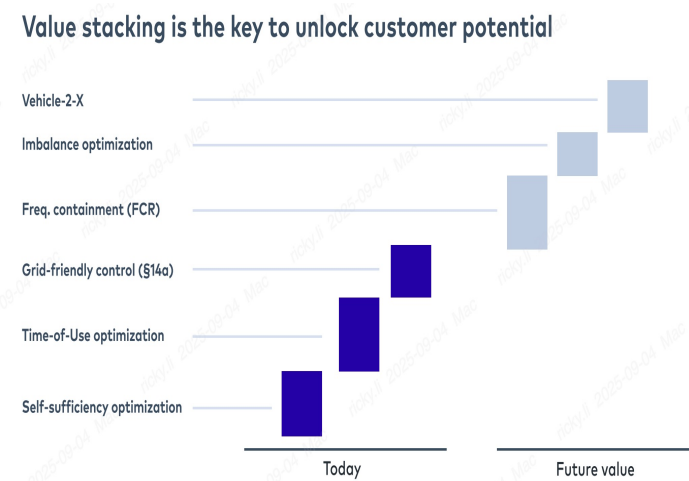
Source: Sigenergy

The absence of standardized scenarios hampers the evaluation of system sizing and energy management algorithms.

Proprietary datasets impede fair and transparent comparisons of vendor performance.

3

Grid Support



Source: gridX

Participation in grid support markets—such as FCR and imbalance optimization—through VPPs can generate new revenue streams.

However, systematic evaluation of residential DERs in VPPs remains limited.



Why open-source?

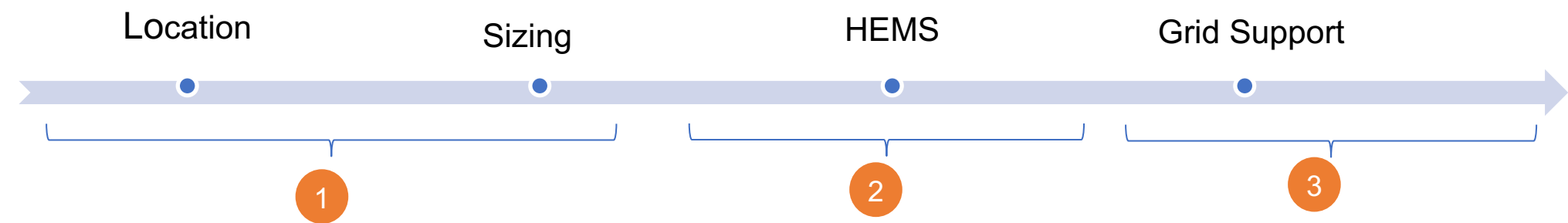
- **Transparent and reproducible benchmarking** for HEMS, PV/ESS sizing, and grid-support strategies.
- **Manufacturer-neutral comparison** through open datasets, scenarios, metrics, and algorithms.
- **Reduced market fragmentation** by creating shared evaluation methods across vendors, researchers, utilities, and end users.
- **Faster ecosystem innovation** by allowing the community to co-develop scenarios, controllers, and validation methods.

Why LF Energy?

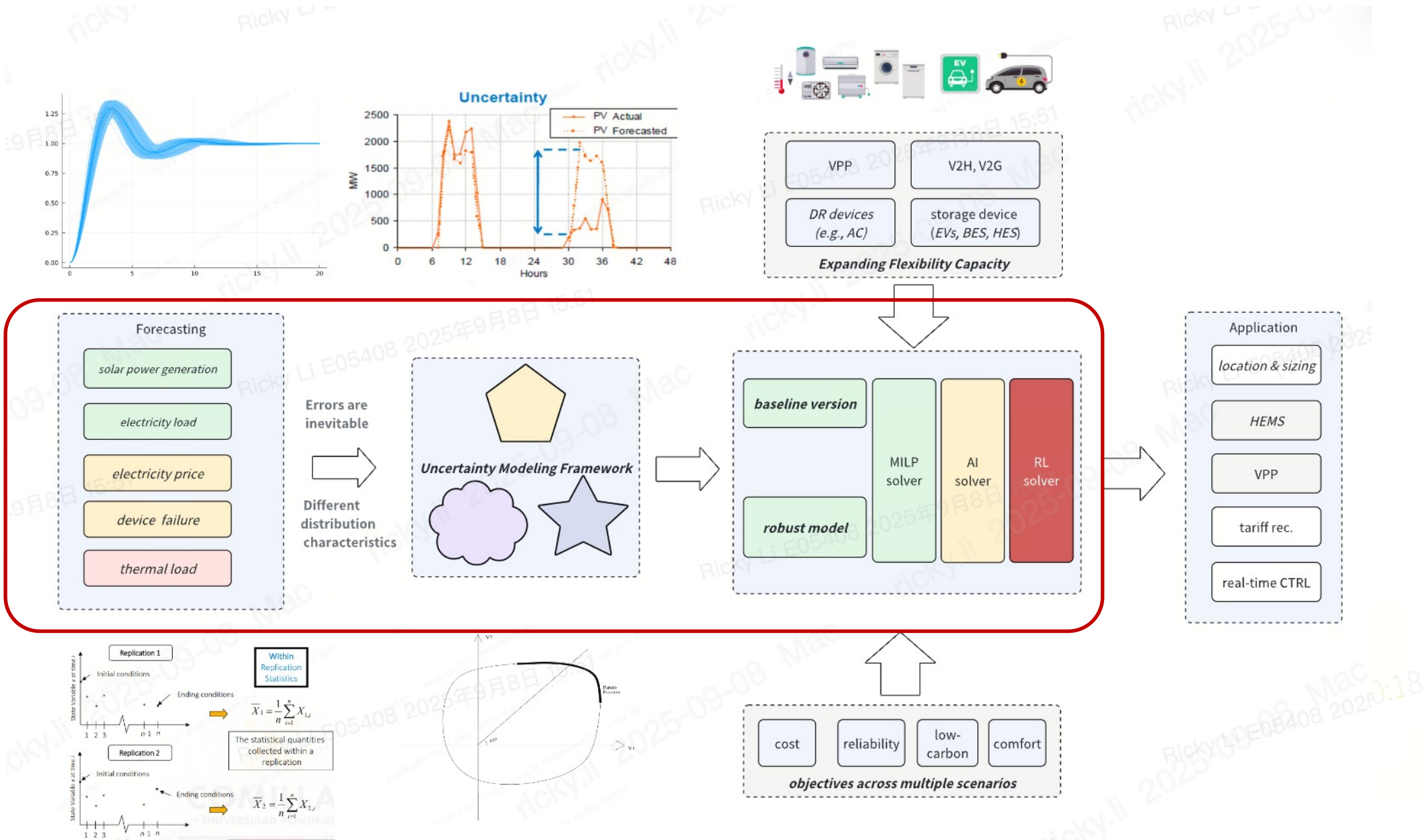
- **Mission alignment:** LF Energy builds shared digital infrastructure for the energy sector through open collaboration, working code, and neutral governance.
- **Neutral governance:** LF Energy provides a trusted home where no single company controls the roadmap.
- **Ecosystem fit:** Smart HEMS Benchmark connects naturally with LF Energy topics such as DER flexibility, grid simulation, interoperability, and AI.
- **Member-driven initiative:** As an LF Energy General Member since 2025, we aim to contribute Smart HEMS Benchmark to support open collaboration across the energy ecosystem.

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Decision-making over the lifecycle of DERs (PV, ESS) can be modeled as an optimization problem with varying decision variables, objectives, and risks, across time horizons from 1 day to 15 years.



Phase	Location & Sizing	HEMS (single-household)	Grid Support
<i>Decision</i>	installation location; capacity	charging/discharging behavior	capacity reserved for VPP
<i>Concern</i>	electricity cost reduction; backup power supply	self-consumption; self-sufficiency; comfort level	additional revenues from grid market
<i>Time horizon</i>	DER lifecycle (~15 years)	typically 1 day	typically 1 year
<i>Risk</i>	accounting for long-term incentive uncertainty	PV generation uncertainty; price mechanism	DER degradation; market scope

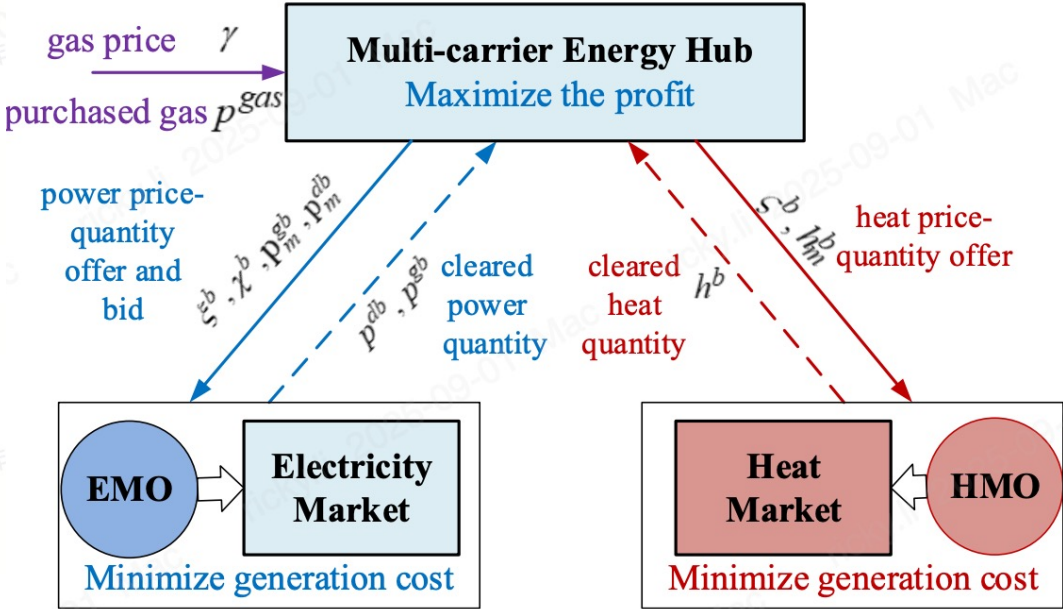


- Example for Location & Sizing & HEMS

- Sample:  ResStock
- 500,000 ResStock load profiles (statistically representative of U.S. households).
- Data Layers:
- Hourly solar generation (PVWatts)
 - Technology costs (NREL)
 - Retail electricity rates (URDB)
 - Local outage events (PRESTO)
- Constraints:
- Battery charge/discharge limits
 - Physical space limitations
 - Cost nonincreasing (affordability)

- Simulation runs:
- Economic viability → max cost reduction
 - Backup viability → max backup power
- Decision variables:
- Whether to install
 - System Sizing
- Output:
- Optimal solar-battery sizing
 - Electricity cost reduction (%)
 - Critical consumption backup (%)

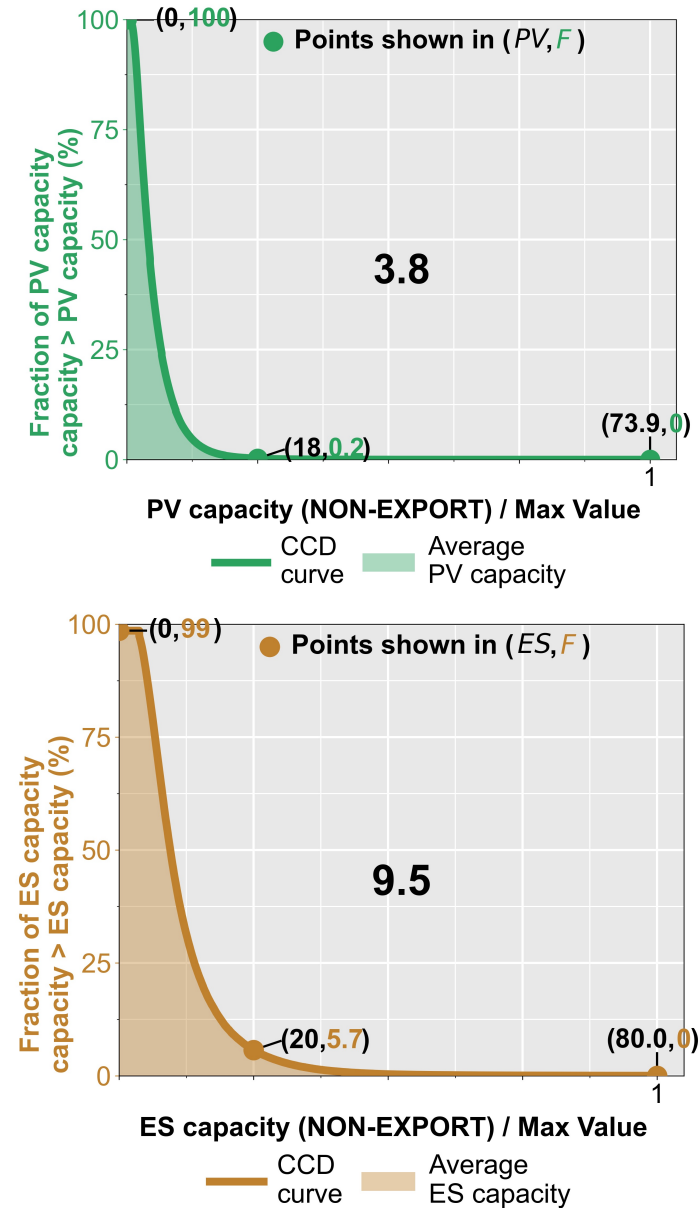
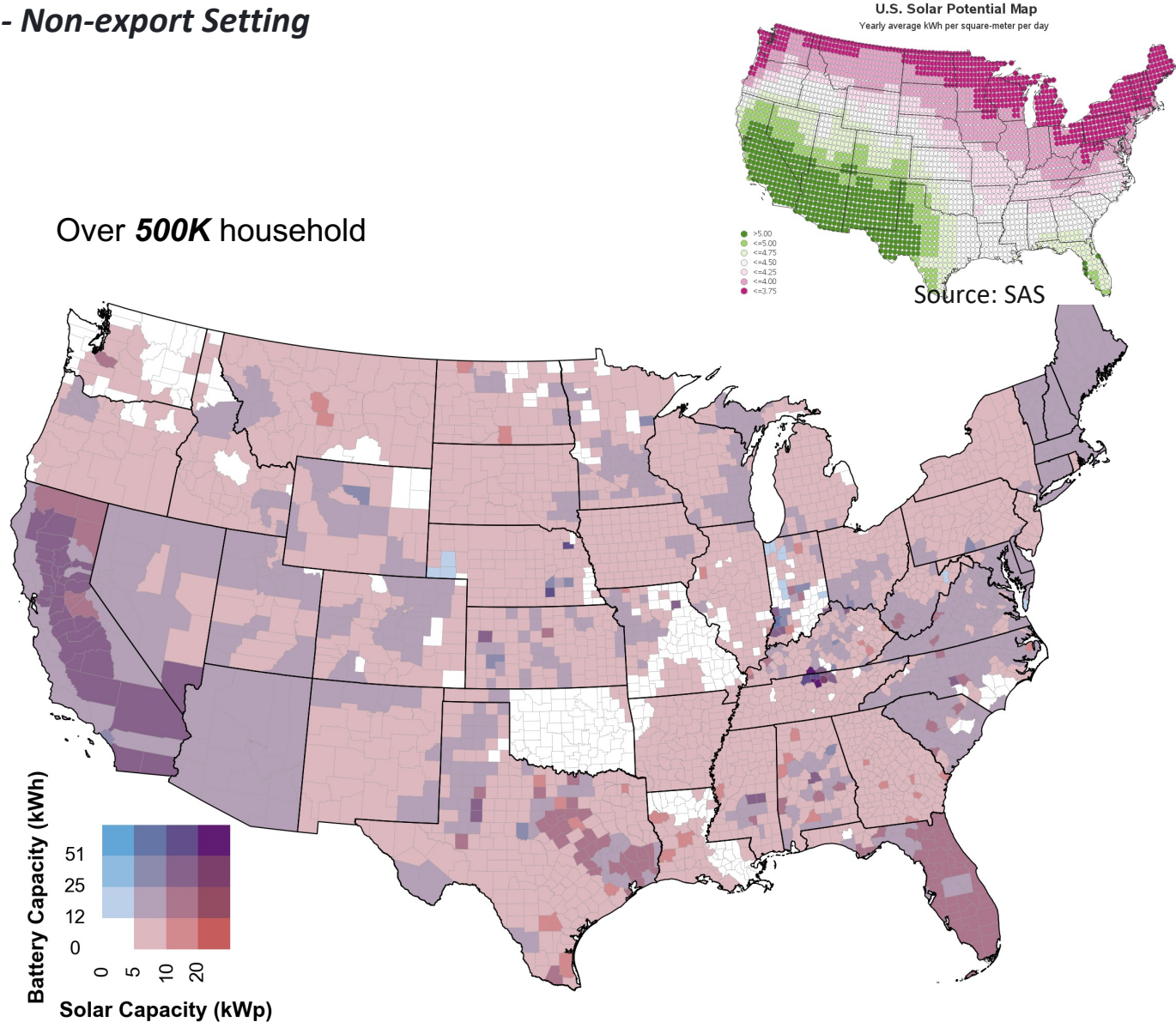
- Example for VPP



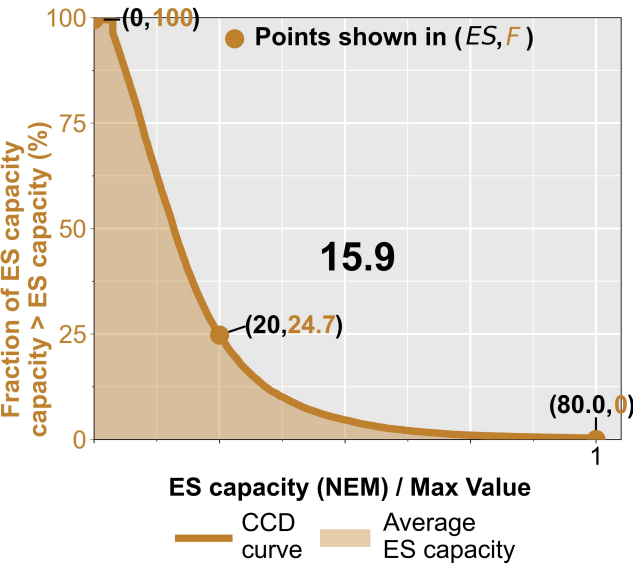
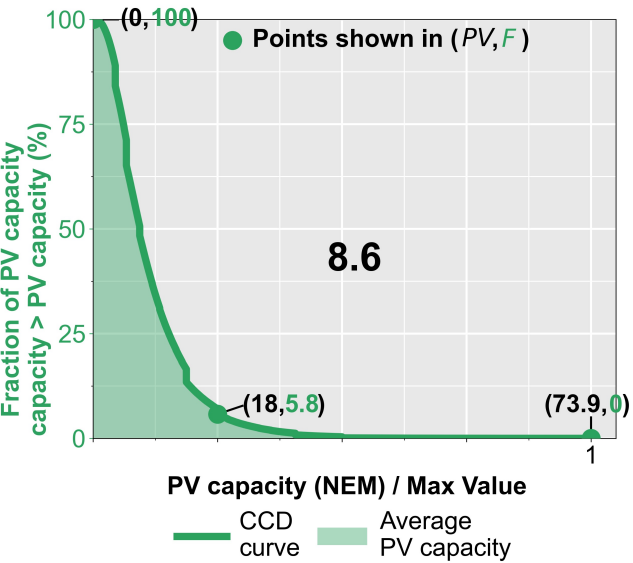
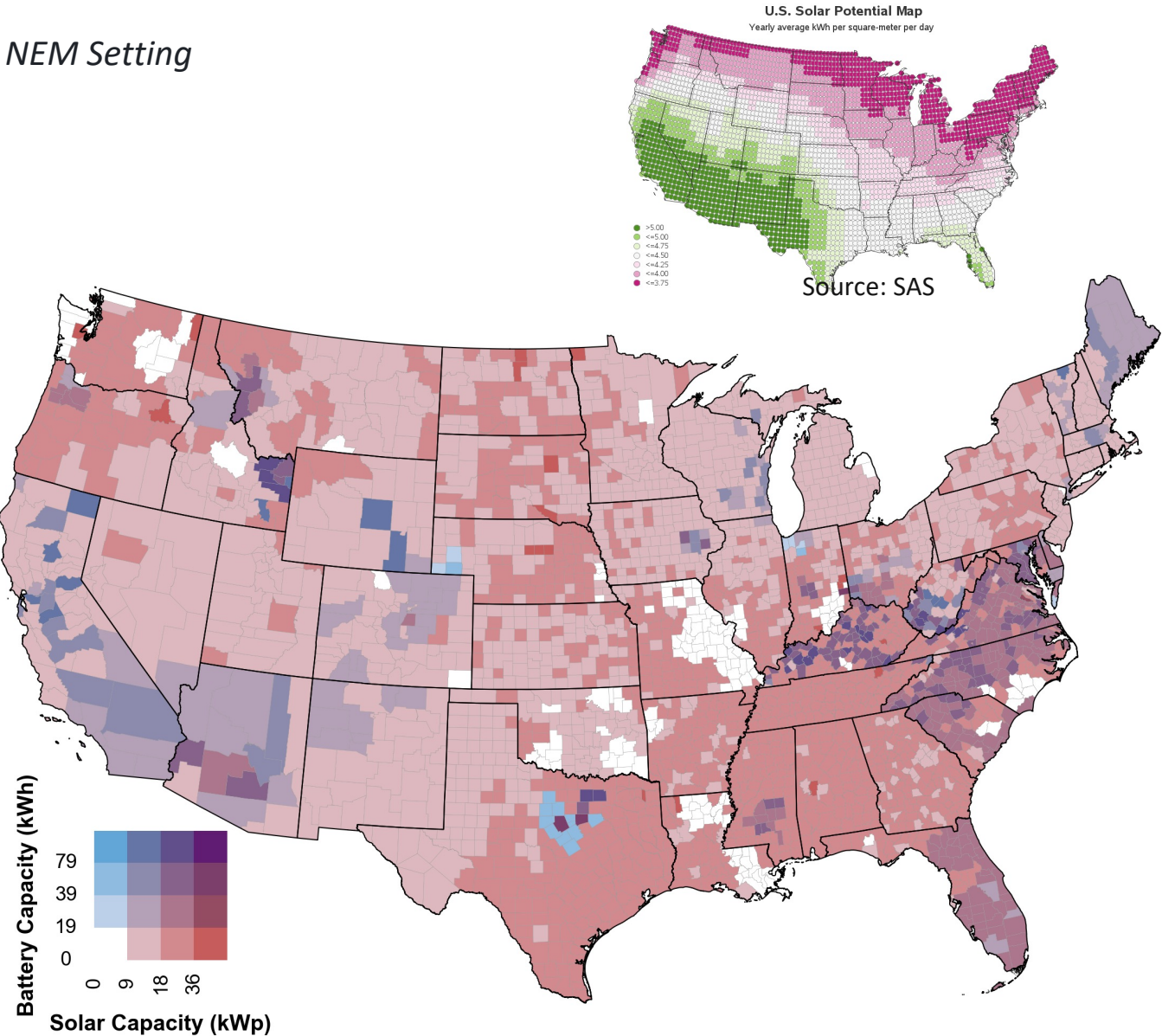
Sun, Tao, et al. "Solar and battery can reduce energy costs and provide affordable outage backup for US households." *Nature Energy* (2025): 1-16.

Li, Rui, et al. "Participation of an energy hub in electricity and heat distribution markets: An MPEC approach." *IEEE Transactions on Smart Grid* 10.4 (2018): 3641-3653.

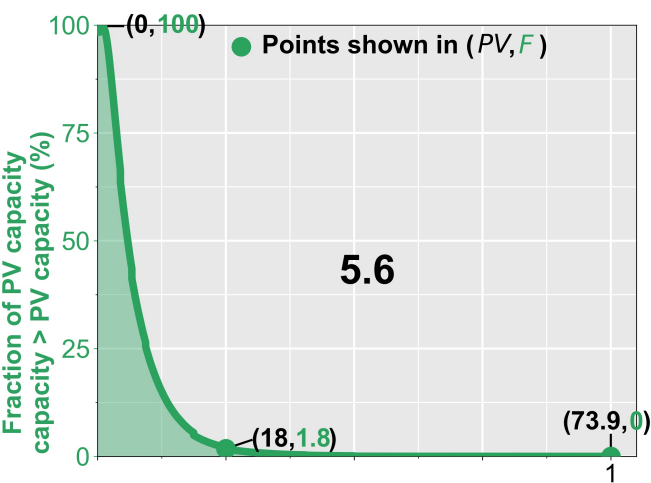
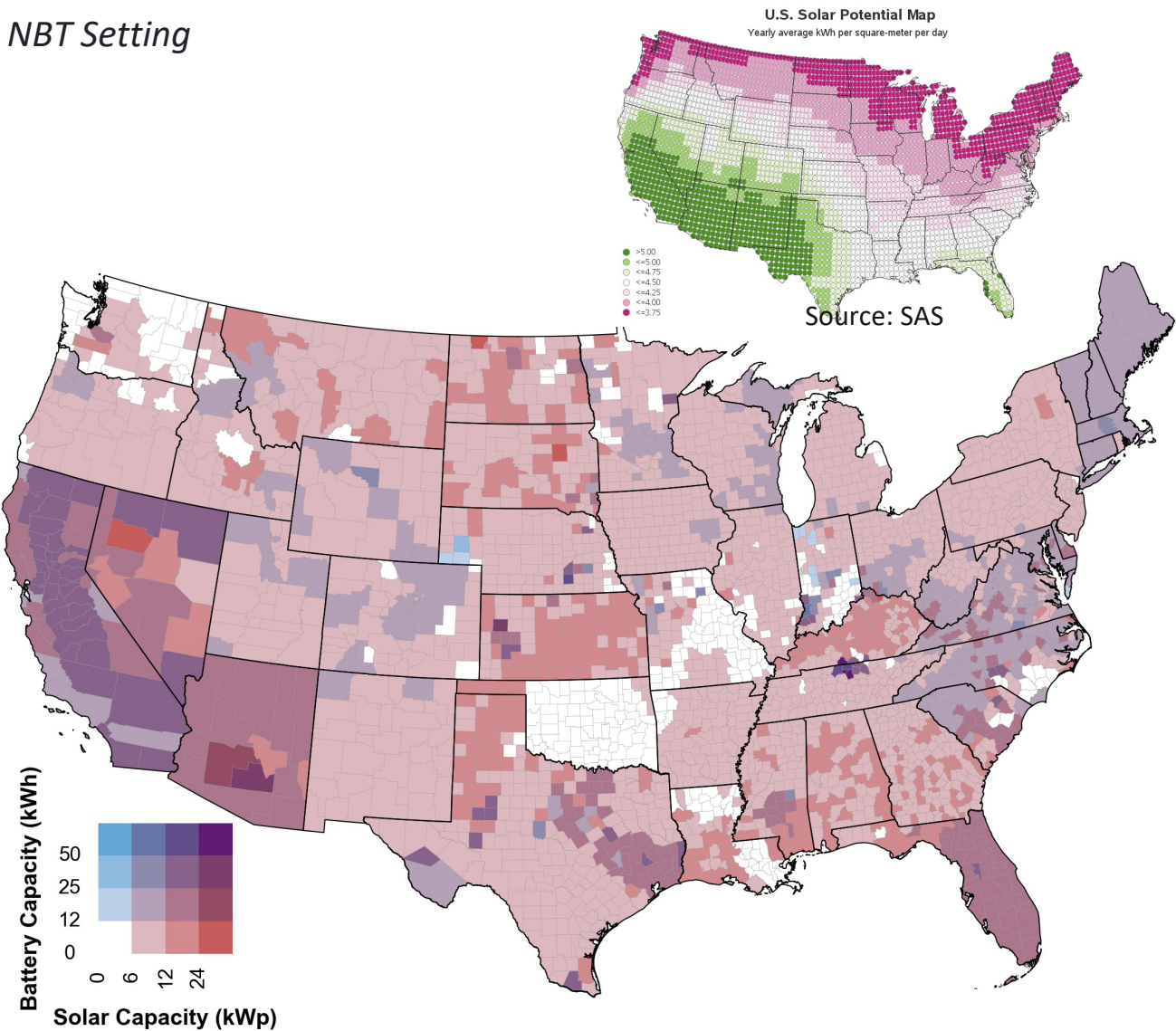
- Non-export Setting



• NEM Setting

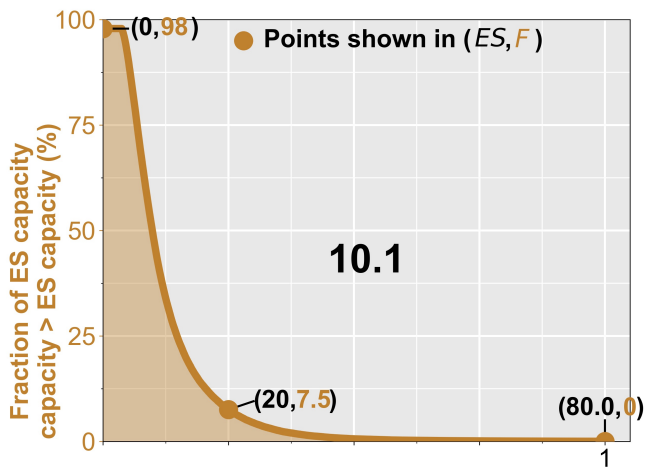


- NBT Setting



CCD curve

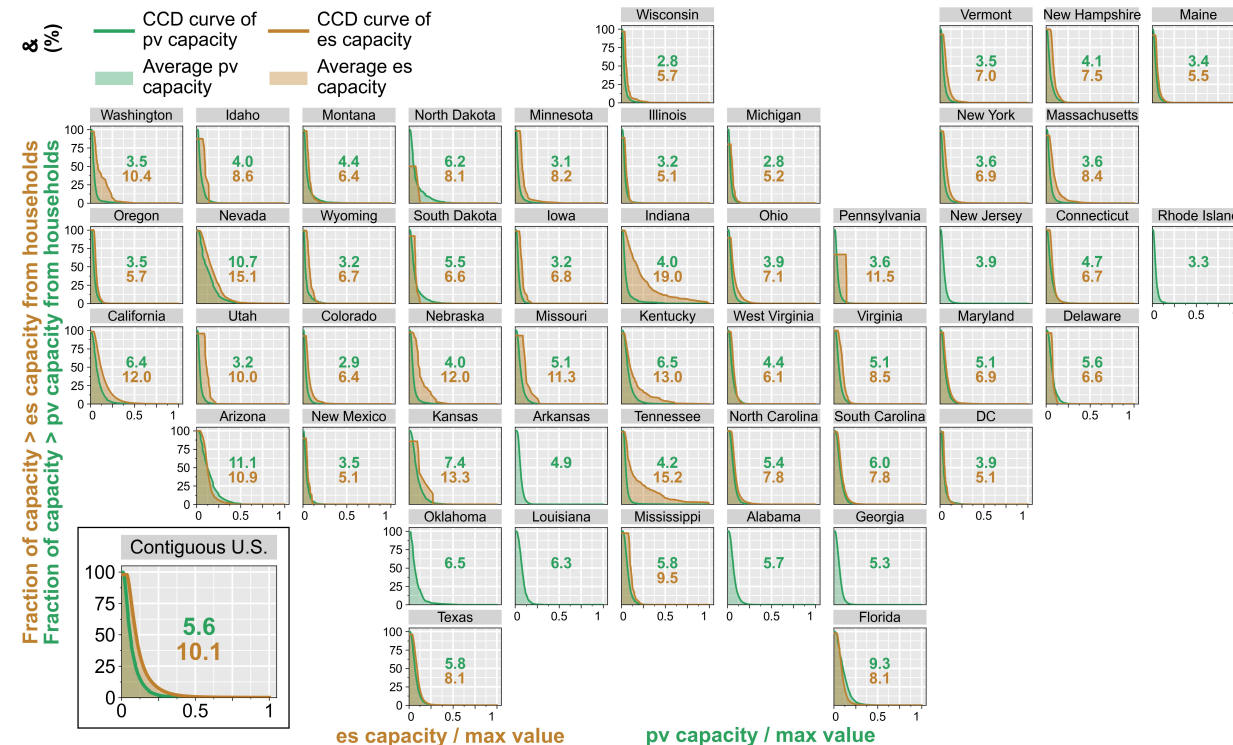
Average PV capacity



CCD curve

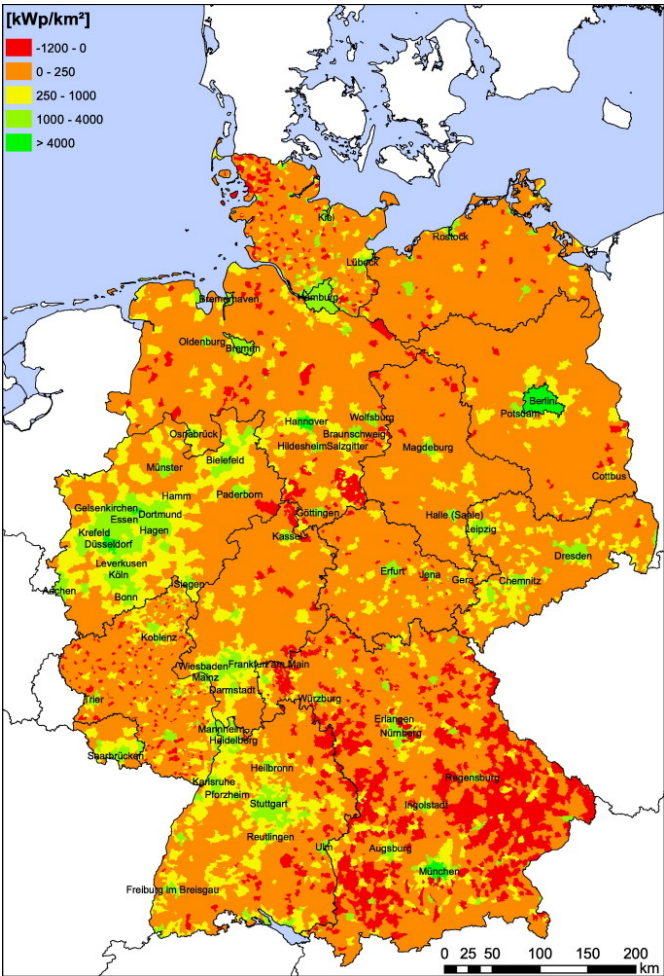
Average ES capacity

-NBT

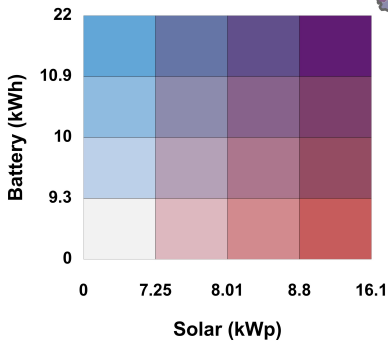


Germany

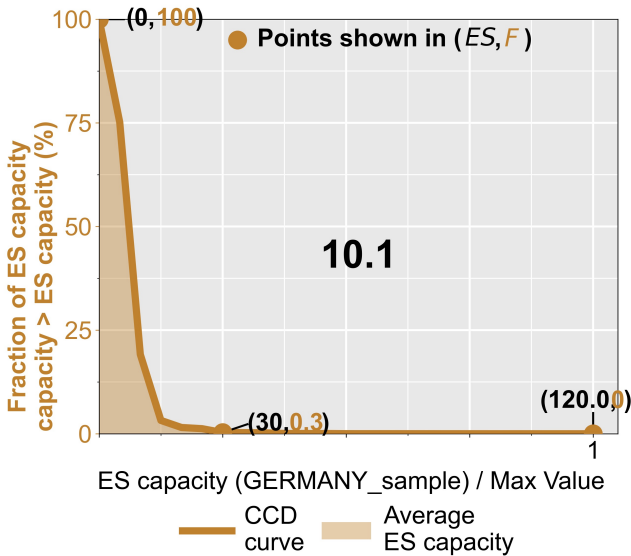
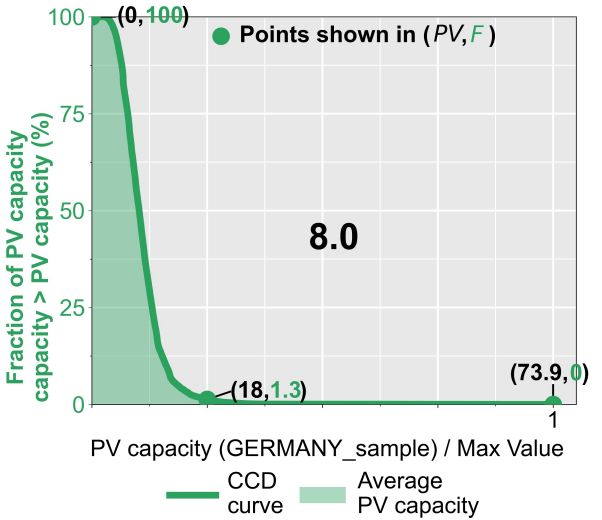
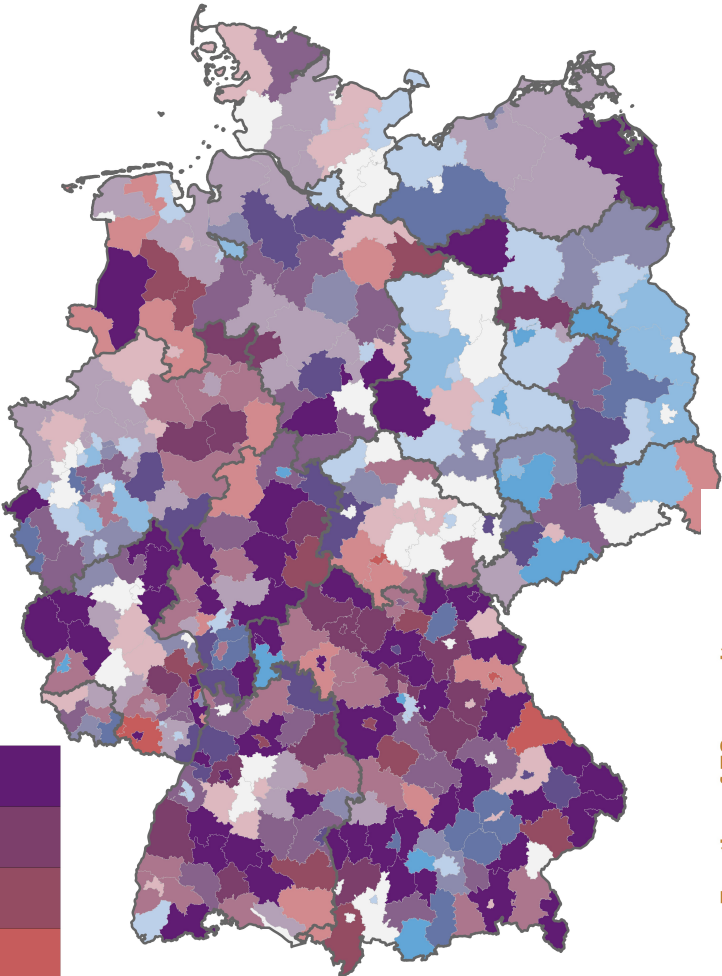
solar resource potential



Source: Solar Energy

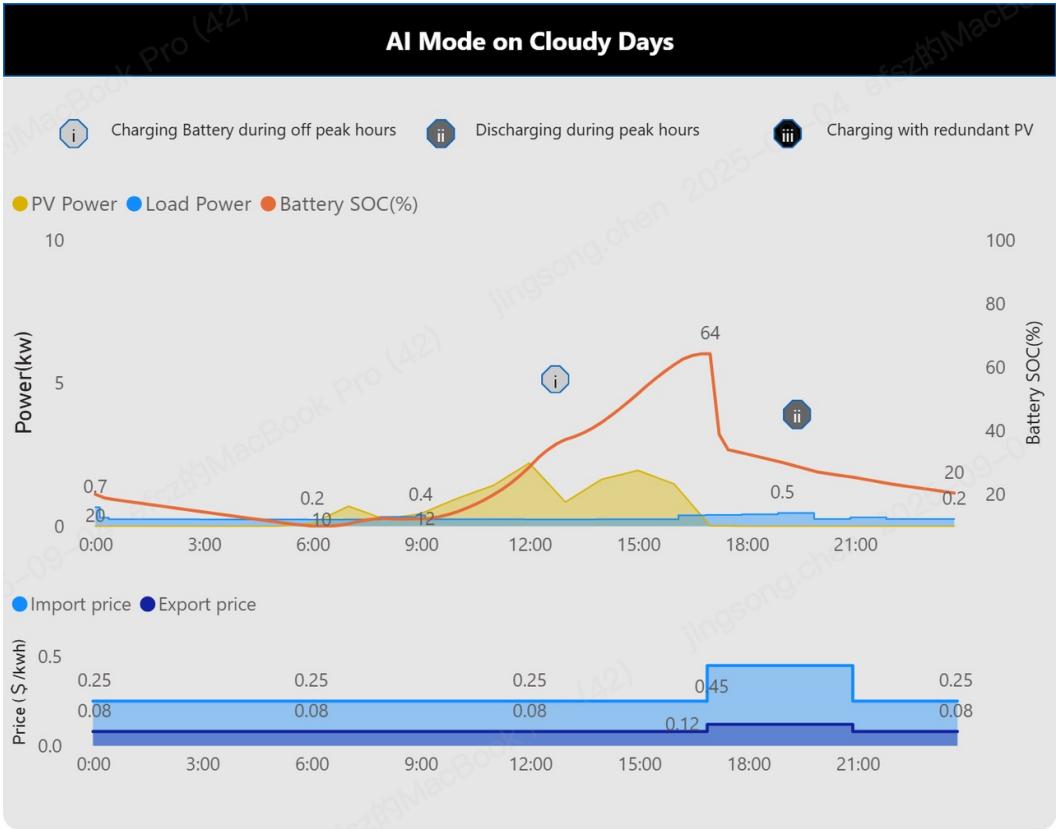
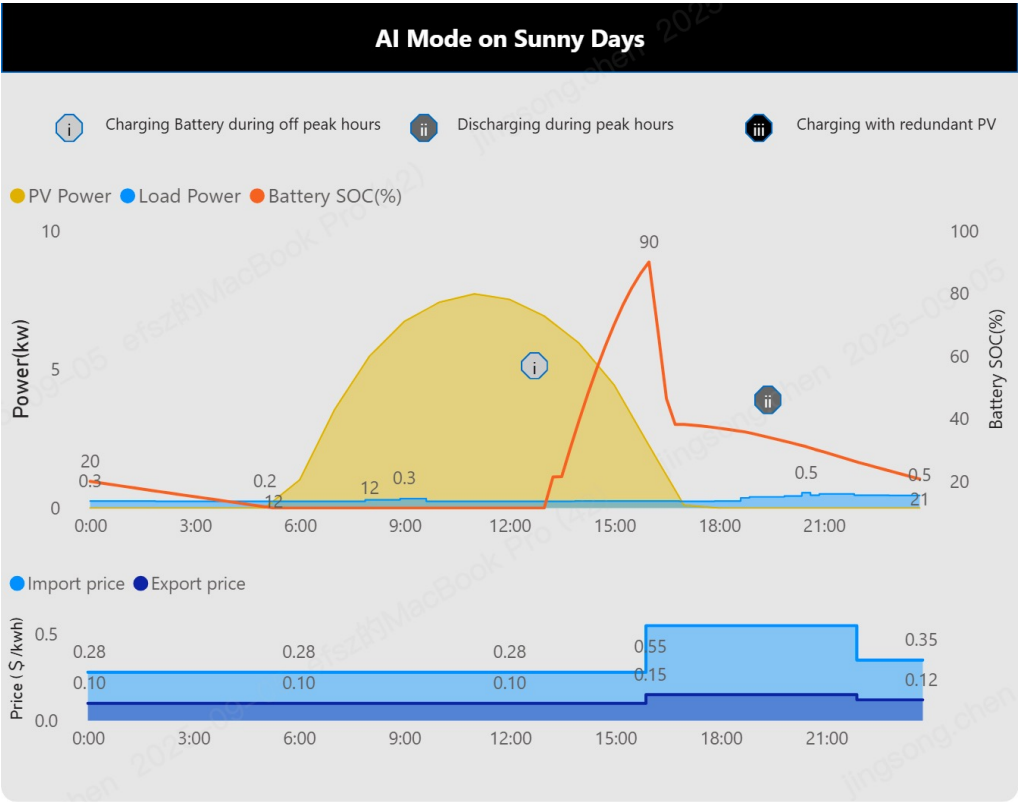


Germany — County-level: Average PV and Storage Capacity per Household



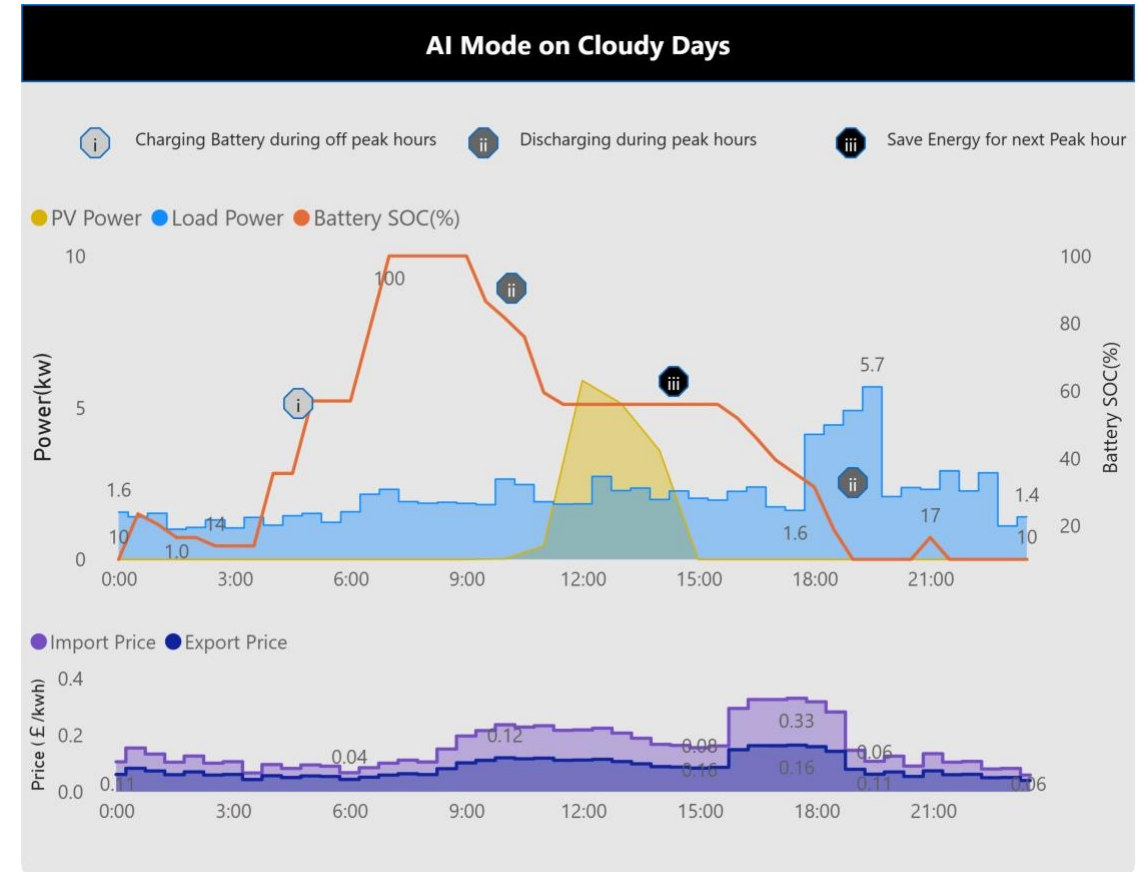
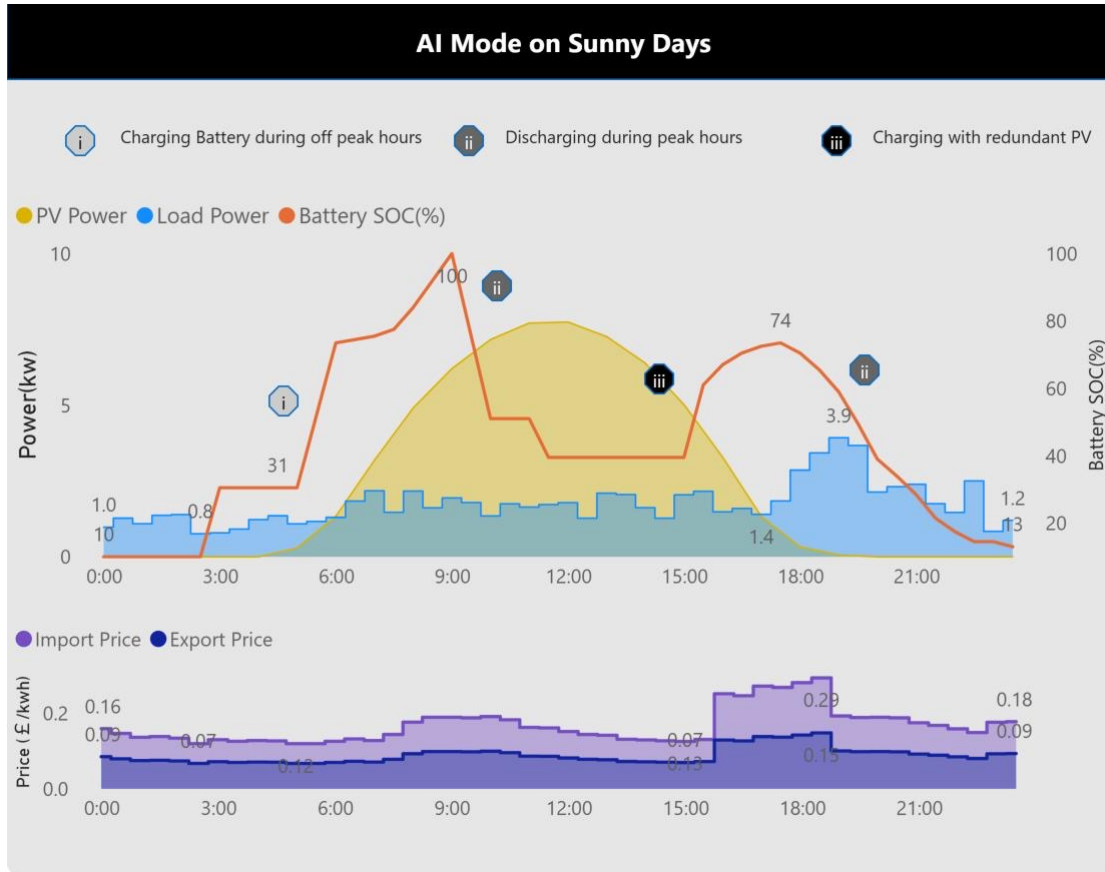
Selected US Case

10kWp PV + 10kWh ESS + TOU



- Selected UK Case

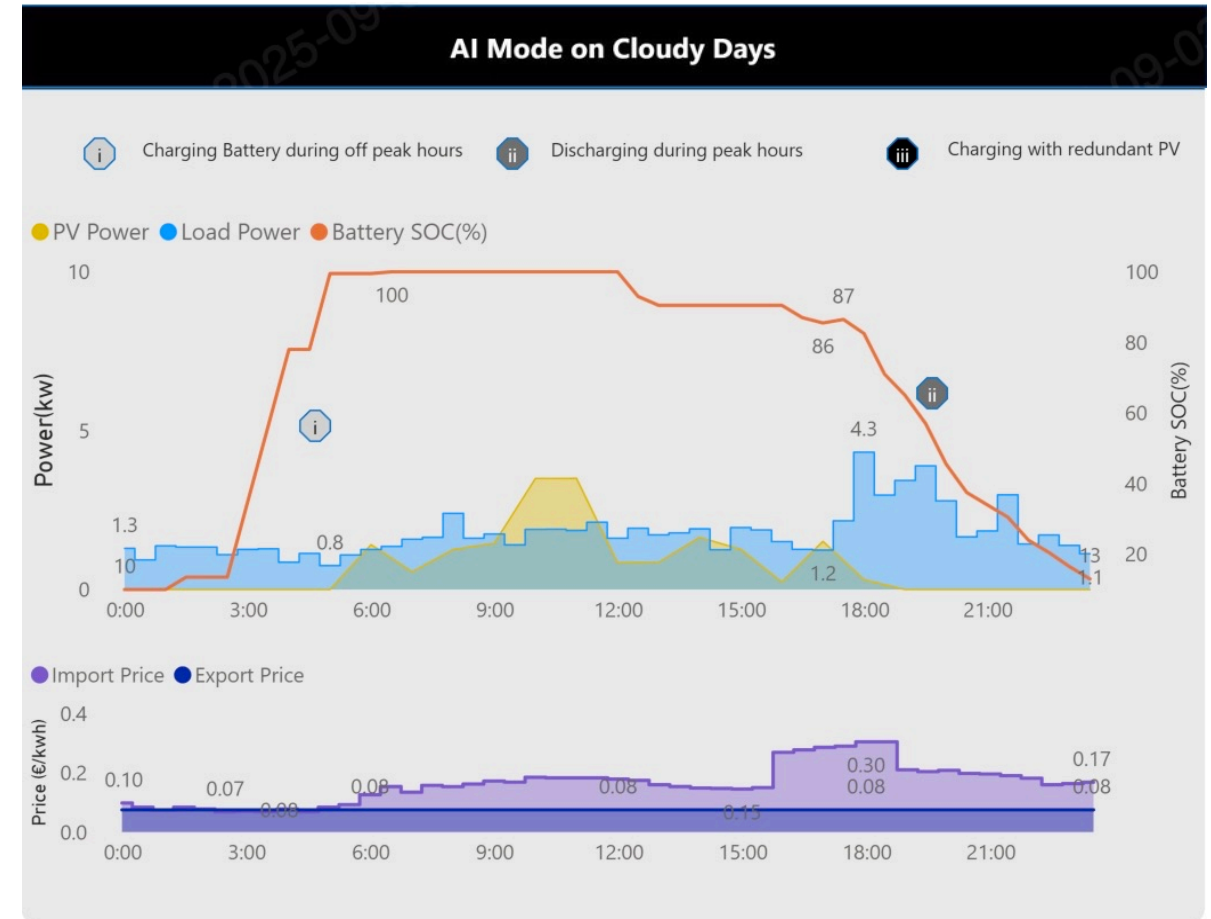
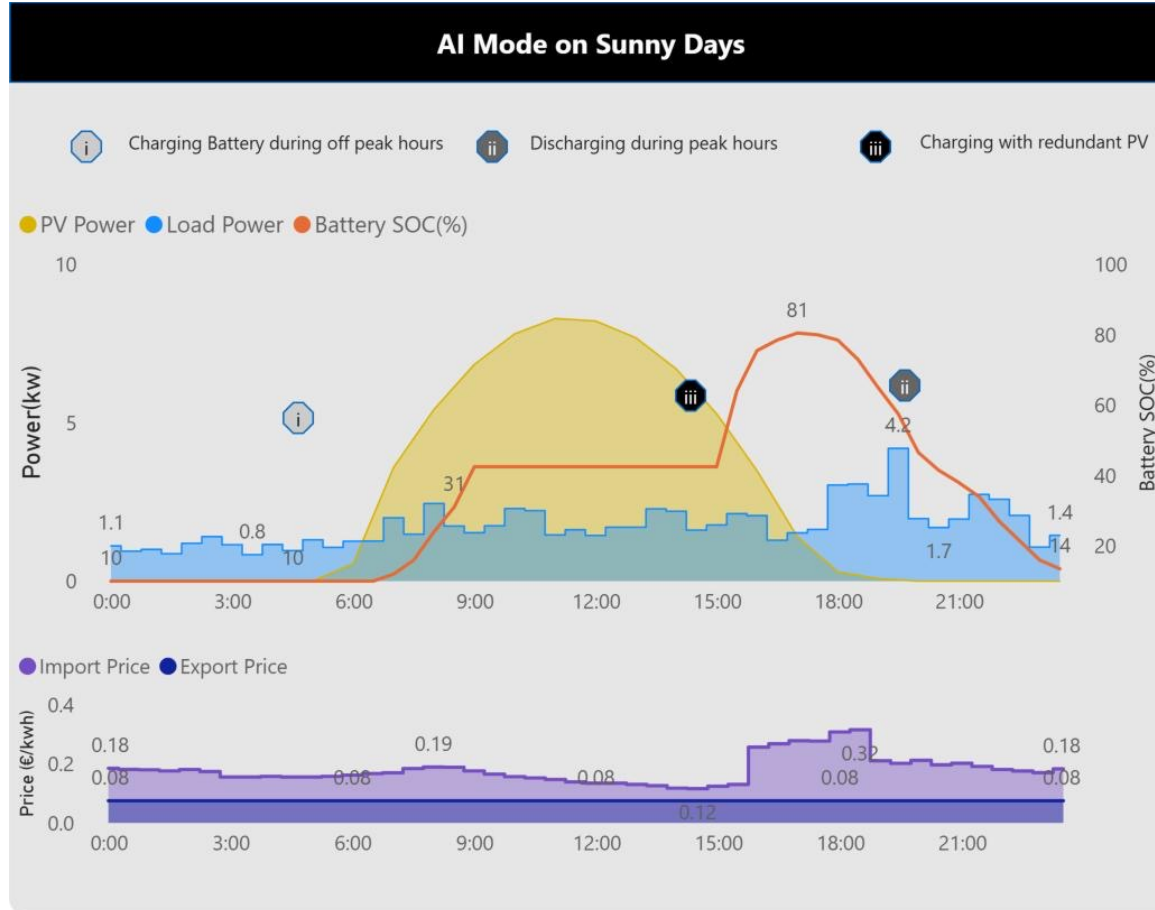
10kWp PV + 20kWh ESS + Octopus dynamic tariff



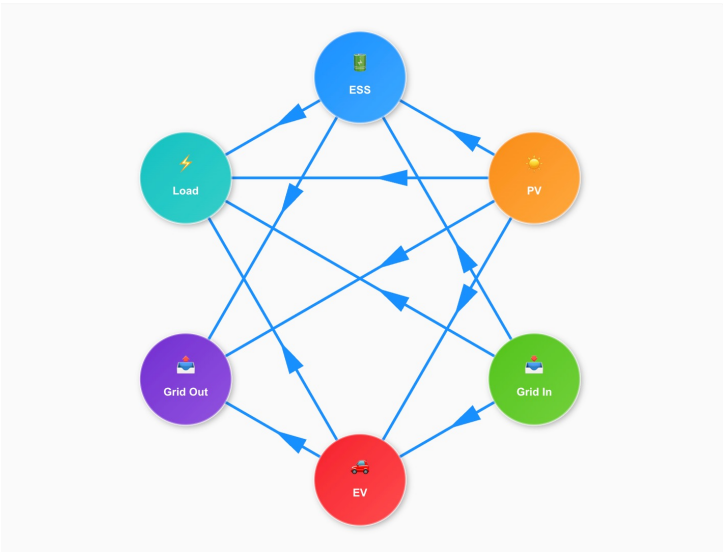
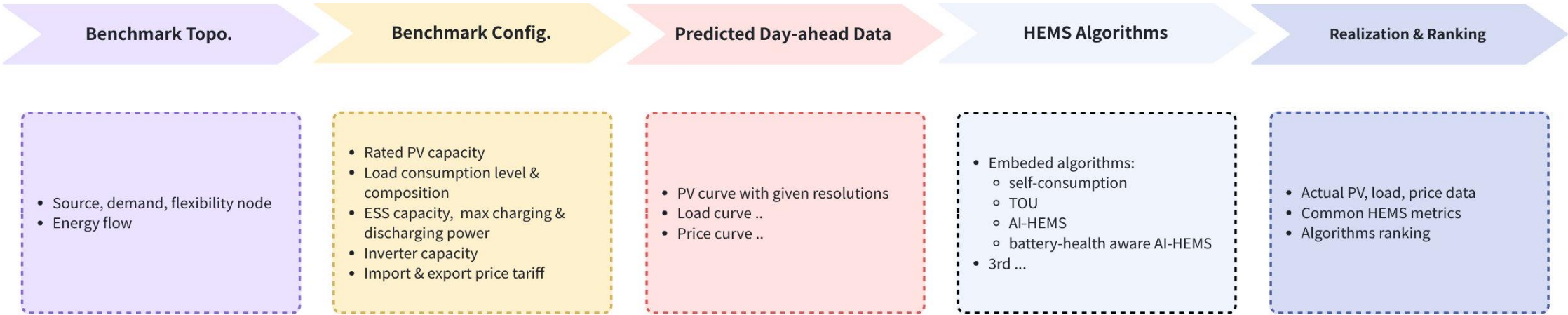
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• Selected Germany Case

10kWp PV + 10kWh ESS + Dynamic tariff



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Country: United States

State: California

City: Los Angeles

PV Baseline Capacity: 4.92 kWp

Expand to configure PV basic system parameters

Expand to configure PV advanced system parameters

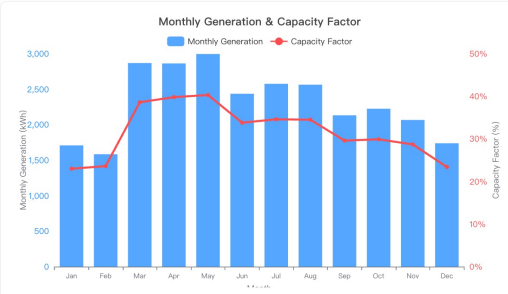
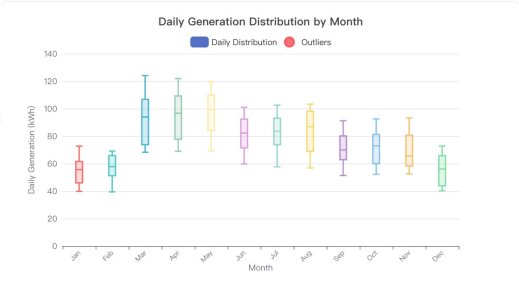
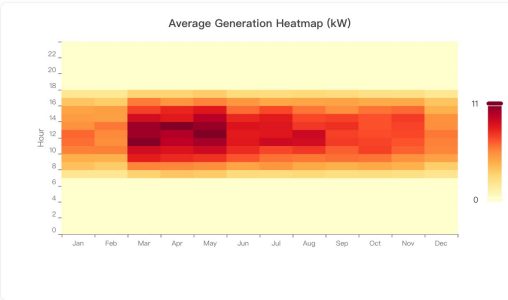
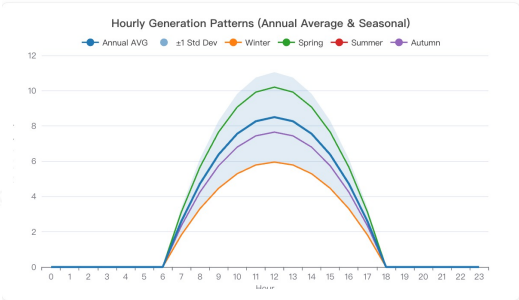
ESS Capacity: 6.93 kWh

Expand to configure ESS system parameters

Daily Power Consumption: 15 kWh

Grid Feed Power: 5.0 kW

Consumption Tariff: te



Universal description language connecting system topology to constraints within HEMS framework.

Benchmark capacity for PV and ESS derived from system location and sizing analyses.

Strategy Selection and Parameter Configuration

Choose the dispatch strategies to be evaluated

Grid-Only

BASIC

Power purchase from grid only, suitable for areas with stable and low grid electricity prices

- Stable Power Supply
- Zero Initial Investment
- Simple Operation

Energy Saving 0%

Cost Optimization 20%

Grid + PV

RECOMMENDED HIGH EFFICIENCY

Grid and PV combination, prioritize PV power during day, use grid power at night

- PV Priority During Day
- Grid Supplement at Night
- Reduce Electricity Bills

Energy Saving 35%

Cost Optimization 45%

Self-Consumption

RECOMMENDED PARAMETER CONFIG HIGH EFFICIENCY

PV power for self-consumption priority, ESS for load regulation, excess power feed-in for revenue

- PV Priority Consumption
- ESS Regulation
- Feed-in Revenue

Energy Saving 60%

Cost Optimization 70%

AI-TOU

RECOMMENDED PREMIUM HIGH EFFICIENCY

AI smart time-of-use price optimization, predict price fluctuations, automatically choose optimal charge/discharge timing

- AI Price Prediction
- Optimal Timing Selection
- Automatic Arbitrage Trading

Energy Saving 50%

Cost Optimization 85%

Robust-X

PREMIUM HIGH EFFICIENCY

Robust optimization strategy, cope with extreme weather and equipment failures, ensure stable system operation

- Extreme Weather Response
- Equipment Fault Tolerance
- Stable Operation Guarantee

Energy Saving 45%

Cost Optimization 80%

Integrated with multiple HEMS algorithms and supports adding new 3rd ones.

- Baseline:

- Grid-only
 - Grid + PV
 - Self-consumption
- Advanced:

- AI-TOU
 - Battery health-aware
 - MPC

Supports sensitivity analysis over:

- PV capacity
- ESS capacity
- Load consumption level
- Grid export limit

Parameter Sensitivity Analysis Configuration

Based on baseline values from step 2, set variation ranges for sensitivity analysis

- Configuration Instructions

 - Default variation range: 80%-120% of baseline
 - Minimum lower bound: 50% of baseline
 - Maximum upper bound: 150% of baseline
 - Minimum step size: 1 for integer parameters

Baseline Review

PV Capacity
4.92 kWp

Daily Consumption
15 kWh

ESS Capacity
6.93 kWh

Grid Power
5 kW

Parameter Configuration

PV Capacity Sensitivity

Enabled

3 ~ 6 kWp

Step 2 kWp

Daily Consumption Sensitivity

Enabled

12 ~ 18 kWh

Step 3 kWh

ESS Capacity Sensitivity

Enabled

5 ~ 9 kWh

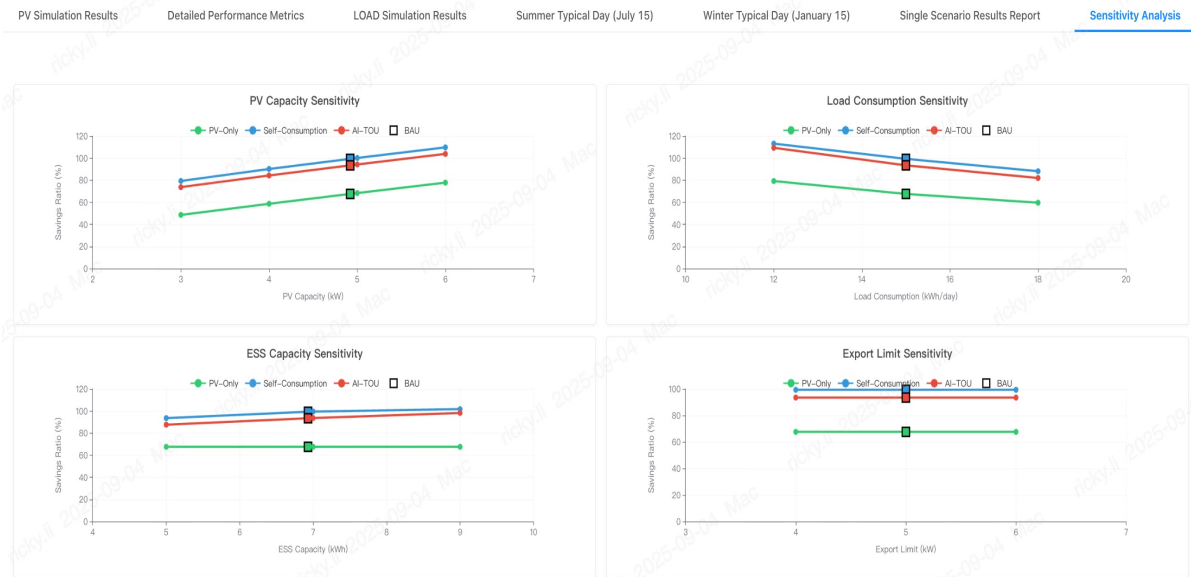
Step 2 kWh

Grid Power Sensitivity

Enabled

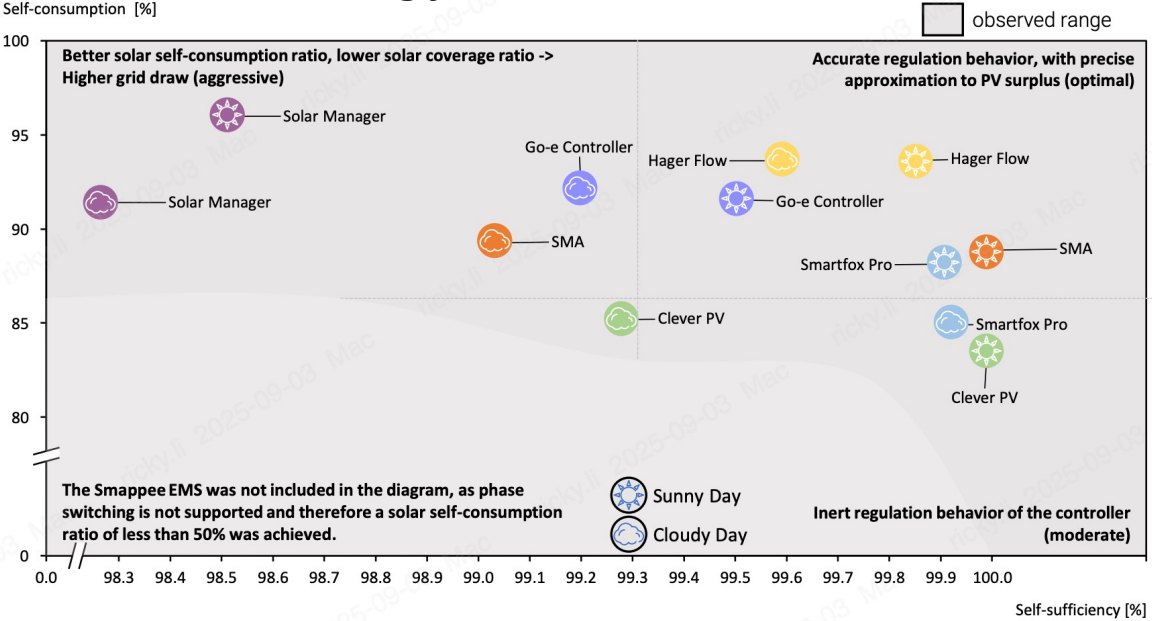
4.0 ~ 6.0 kW

Step 1 kW



month	daily load (kWh)	pv generation (kWh)	ess capacity (kWh)	original cost (\$)	cost with pv (\$)	cost with pv-ess & self-consumption (\$)	cost with pv-ess & ai-tou (\$)	pv w.r.t original (%)	pv-ess w.r.t original (%)	ai-tou w.r.t original (%)
2024-01	14.7	13.3	6	59.91	-19.1	-19.18	-33.64	131.9	132	156.1
2024-02	11	17.7	6	46.61	-32.14	-32.12	-45.57	169	168.9	197.8
2024-03	7.1	24.4	6	46.44	-44.19	-43.42	-58.05	195.1	193.5	225
2024-04	6.1	31	6	41.5	-53.08	-50.98	-67.06	227.9	222.9	261.6
2024-05	7.1	35.4	6	47.6	-54.5	-51.04	-71.08	214.5	207.2	249.3
2024-06	15	37.6	6	57.79	-47.46	-47.85	-62.6	182.1	182.8	208.3
2024-07	15	39.9	6	90.57	-21.85	-22.08	-38.44	124.1	124.4	142.4
2024-08	19.1	35.4	6	90.33	-28.84	-28.74	-45.02	131.9	131.8	149.8
2024-09	15	28.8	6	75.03	-34.95	-35.28	-50.5	146.6	147	167.3
2024-10	23.5	22.1	6	67.06	-37.4	-37.33	-53.99	155.8	155.7	180.5
2024-11	23.1	15.5	6	53.9	-28.61	-28.38	-42.25	153.1	152.7	178.4
2024-12	19.1	11.1	6	45.94	-25.31	-25.19	-39.51	155.1	154.8	186
SUM(AVG)				722.68	-427.43	-421.59	-607.71	165.59 (AVG)	164.47 (AVG)	191.88 (AVG)

Two-metric ranking from P3



Source: P3 Group

Collaboration with **HEMS Finder** is under discussion for conducting a multi-metric ranking of available HEMS on that platform.

- Metrics:
 - Cost & revenue
 - Comfort level
 - Battery degradation & LOCE



- **Current users:** The project is currently used internally by HEMS-related development and product teams at EcoFlow.
- **Early external interest:** Several LF Energy-related project participants and HEMS domain experts have already expressed interest, including researcher from Lawrence Berkeley National Laboratory.
- **LF Energy community exposure:** We presented the project at the 2025 LF Energy European Summit and have also submitted a demo proposal for this year's Summit to showcase the project live.
- **Academic outreach:** The project initiators have strong academic backgrounds across Tsinghua, Stanford, and TUM, enabling outreach through research communities in China, the US, and Europe.

- **Open-source engagement:** Ricky has contributed to and maintained open-source energy software on GitHub since 2014, building connections with energy researchers, faculty members, and developers.
- **European HEMS testing network:** TUM CoSES operates a HEMS PHIL testing platform in Europe and hosts an annual HEMS Symposium in September, providing a natural channel to engage researchers, vendors, and testbed operators.
- **Industry platform adoption:** HEMS Finder, a third-party platform with 100+ HEMS service providers, has expressed interest in adopting Smart HEMS Benchmark as part of its HEMS evaluation platform.

- **Initial committers:** The initial committer base will come from the EcoFlow initiation team, HEMS-related labs at Tsinghua, Stanford, and TUM, and Ricky's open-source GitHub network, providing early coverage across China, the US, and Europe.
- **Convert early users into contributors:** We plan to engage interested LF Energy project participants, HEMS domain experts, and early benchmark users to contribute datasets, scenarios, algorithms, documentation, and test cases.
- **Challenge-based contributor onboarding:** During the HEMS Symposium, we plan to co-organize benchmark challenges with HEMS Finder to attract developers from HEMS service companies as well as students and researchers.
- **Industry and academic contributor pipeline:** HEMS Finder's network of 100+ HEMS service providers, together with TUM CoSES's HEMS PHIL platform and academic outreach channels, can help bring in both industry developers and research contributors.
- **LF Energy community engagement:** If accepted, we will use LF Energy channels, SIGs, summit demos, and related project connections to attract contributors and align Smart HEMS Benchmark with the broader LF Energy ecosystem.